

Nitik Jain

Robotics Researcher & Applied-AI Engineer

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Summary

Robotics researcher and applied-AI engineer pursuing an M.S.E. in Robotics at Johns Hopkins University, with ~3 years of full-time experience shipping production AI across medical imaging, fintech, and autonomous driving. Research spans robot learning, vision-language-action models, perception, and generative & medical AI. Seeking **Applied Research** roles and high-impact **early-stage teams** where research depth meets real-world shipping.

Education

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| <p>MSE Johns Hopkins University, Robotics (Perception & Cognitive Systems)</p> <ul style="list-style-type: none"> Student Researcher, Johns Hopkins Medicine — deep learning & AI for MRI enhancement and segmentation. Coursework: Computer Vision, Deep Learning, Algorithms for Sensor-Based Robotics, Generative Models for Vision. | <p>Baltimore, MD, USA
Aug 2025 – May 2027</p> |
| <p>MTech Indian Institute of Technology (IIT) Kanpur, Aerospace Engineering</p> <ul style="list-style-type: none"> GPA: 9.4 / 10 — Academic Excellence Award 2021 (Best Academic Performance). Master's Thesis: <i>Object Detection Underneath Sparse Occlusion of Forests</i> (Advisor: Dr. Mangal Kothari). | <p>Kanpur, India
July 2021 – Aug 2022</p> |
| <p>BTech Indian Institute of Technology (IIT) Kanpur, Mechanical Engineering (Minor in Machine Learning)</p> <ul style="list-style-type: none"> GPA: 7.4 / 10 — Minor in Machine Learning. Lead AI Researcher, Team VISION (autonomous ground vehicles). | <p>Kanpur, India
July 2017 – June 2021</p> |

Experience

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| <p>Swiffylabs.ai, Lead AI Engineer</p> <p>WestBridge Capital-backed B2B fintech redefining core banking through applied AI; led AI development from 0 to 1.</p> <ul style="list-style-type: none"> Designed and deployed an AI-driven core-banking stack, replacing legacy systems with modular microservices. Integrated LLM agents and fraud-detection models into transaction, onboarding, and customer-experience workflows. Partnered with NBFC clients to launch a regulatory-ready, headless product suite, leading the first commercial rollout. | <p>Bangalore, India
June 2024 – June 2025
1 year 1 month</p> |
| <p>Osfin.ai, Founding AI Engineer</p> <p>Led product vision for LLM-powered document analysis and summarization.</p> <ul style="list-style-type: none"> Led cross-functional teams to define and implement strategic product roadmaps as a founding engineer. Directed an AI research team building LangChain-based products from the ground up. Designed OCR-based invoice extraction in Bahasa, improving accuracy by 40%. Built an LLM-based email parser spanning 30+ categories and a clustered Vector-DB ETL pipeline that cut search space by 80%. | <p>Bangalore, India
Apr 2023 – June 2024
1 year 3 months</p> |

Qure.ai, AI Scientist, R&D Division

Deep-learning 3D computer-vision research on qCT-Chest, Qure.ai's flagship AI lung-cancer detection suite.

- Developed deep-learning algorithms to detect early-stage cancer from lung CT scans.
- Implemented a 3D ViT in MONAI, achieving classification AUC 88% and recall 96% (+9% over SoTA).
- Integrated CBAM and SE layers into a U-Net backbone, improving nodule-segmentation DICE by 14%.
- Streamlined training with an end-to-end YAML framework on PyTorch Lightning + Hydra.

Bangalore, India
Sept 2022 – Mar 2023
7 months

MooVita Pte Ltd, AI Research Consultant

Perception researcher in the computer-vision division for an autonomous-driving suite.

- Leveraged 2D features to build a 3D sparse point cloud localized on monocular odometry.
- Engineered a classical optical-flow fallback for rapid vehicle-turning estimation at 240 FPS on CPU.
- Devised a fallback object-detection pipeline exploiting RAFT dense optical flow.

Singapore
Sept 2022 – Feb 2023
6 months

Unilever Limited, ULIP Intern, Engineering Team

Among 17 students selected pan-India for the prestigious Unilever Leadership Internship Programme.

- Evaluated and recommended a warehouse-automation work-stream, saving 8,000+ man-hours (~₹1 Cr/yr).
- Awarded a pre-placement interview based on internship performance.

Mumbai, India
Apr 2020 – June 2020
3 months

Research Experience

IGCL, IIT Kanpur, Sr. Student Associate Researcher

Master's Thesis: Object Detection Underneath Sparse Occlusion of Forests (Advisor: Dr. Mangal Kothari).

- Built a sensor-fusion model integrating RGB and thermal modalities for human detection in forests.
- Implemented Airborne Optical Sectioning (AOS) synthetic-aperture imaging, achieving ~70% occlusion removal.
- Fine-tuned YOLOv5 via transfer learning for a 32% increase in detection accuracy.

Kanpur, India
June 2021 – Aug 2022

Robotics Research Center, IIIT Hyderabad, Visiting Research Intern

Motion Planning for Outdoor Autonomous UAVs (Advisor: Prof. K. Madhava Krishna).

- Implemented VINS-Mono on ROS, fusing monocular camera and IMU for self-localization.
- Integrated GPS-RTK telemetry with VINS odometry and achieved real-time planning via mav_voxblox_planning.

Hyderabad, India
May 2019 – July 2019

Team VISION, IIT Kanpur, Lead AI Researcher

Autonomous Ground Vehicle — Daksh II (Advisor: Dr. Mangal Kothari).

- Implemented Fast-SCNN semantic segmentation with MobileNet depth-wise separable convolutions.
- Fused LiDAR + camera for occupancy mapping and used costmap_2d with RRT for cubic-spline path planning.
- Won 4 accolades at IGVC-2019; published at IEEE ICARSC 2022.

Kanpur, India
Aug 2018 – May 2020

Publications

Multimodal Object Detection Under Sparse Forest-Canopy Occlusion

Nitik Jain, Mangal Kothari

(Preprint (extends M.Tech thesis, IIT Kanpur))

2026

IR Models and the COVID-19 Pandemic: A Comparative Study of Performance and Challenges

2023

M. Shukla, Nitik Jain, S. Gupta
(arXiv:2305.12528)

Development of a Low-Cost Autonomous Ground Vehicle

Apr 2022

Nitik Jain, A. A. Shah, H. Bollamreddi, Mangal Kothari
[10.1109/ICARSC55462.2022.9784787](https://doi.org/10.1109/ICARSC55462.2022.9784787) (IEEE ICARSC 2022)

Lightweight Multi-Drone Detection and 3D-Localization via YOLO

2022

A. Sharma, Nitik Jain, Mangal Kothari
(arXiv preprint)

Patents

1. *Computer-Implemented Stigma Separation System* — Indian Patent No. 564565 (filed as App. No. 202311082765), 2025. Commercialized with the University of Kashmir.

Honors & Awards

- **J. N. Tata Scholar** (2025) — one of ~100 scholars worldwide for postgraduate study abroad.
- **K. C. Mahindra Scholar** (2025) — one of ~90 scholars worldwide for postgraduate education overseas.
- **Lescoe Cup, 2nd Worldwide & 3rd, AutoNav Challenge** — 27th Intelligent Ground Vehicle Competition, Michigan (2019), among 37 international institutes.
- **SIIC Student Innovation Award**, IIT Kanpur (2022) — for the Saffron Stigma Separator.
- **Academic Excellence Award**, IIT Kanpur (2022) — exceptional academic performance in the dual-degree program.
- **Silver Medal**, 7th Inter-IIT Tech Meet (2019) — TCTD Challenge, Tata Centre for Technology & Design.
- **NVIDIA Hardware Academic Grant** (2020, GPUs worth \$3,000) & **AWS Machine Learning Scholarship** (2021).
- **JEE Advanced — All-India Rank 1116** (2017), among ~1.5 million candidates.

Leadership & Entrepreneurship

Team VISION (ex-IGVC), IIT Kanpur, Team Leader

Kanpur, India

- Spearheaded a three-tier, 21-person team driving AI & robotics research.
- Raised \$35,000 in funding and secured the NVIDIA Hardware Academic Grant.
- Led the team to 4 international awards at IGVC-2019 (Michigan) among 37+ teams.

June 2019 – May 2021

Tralee.ai, Co-Founder

IIT Kanpur

SkipQ — an AI solution to automate checkouts at offline retail stores.

June 2020 – Feb 2021

- Secured \$12,000 pre-seed ideation investment from the IITK-MIAP program.
- National Finalist (Top 10 of 1,600+ entries, Prabandhan-MBA); Asia Finalist at the La Trobe Tech Infusion Challenge.

Teaching & Service

Teaching Assistant: Innovative Data Strategy for Global Leaders, Johns Hopkins University (Spring 2026, Prof. Shadi Esnaashari).

Teaching Assistant: AE640A: Autonomous Systems, IIT Kanpur (Spring 2022, graduate course).

Invited Talk: ML & Computer Vision tutorial, QIP Short-Term Course on Decision & Control, IIT Kanpur (2021).

Paper Reviewer: 7th Indian Control Conference (ICC-7), 2021.

Open Source: Google Summer of Code — INCF (2021, bio-robotics sensor fusion) and ML4SCI (2026, latent diffusion for physics).

Skills

ML / Deep Learning: PyTorch, TensorFlow, MONAI, Hugging Face, LangChain, Computer Vision, NLP, 3D Vision

Robotics & Perception: ROS, SLAM, Sensor Fusion, Object Detection, Semantic Segmentation, YOLO, Visual Odometry

Programming: Python, C++, CUDA, MATLAB, SQL, Git

Cloud & MLOps: AWS, Docker, Kubernetes, Databricks, MLflow, Hydra, PyTorch Lightning